

Project Plan – MLOps Deployment for Plant Phenotyping

Introduction

This project builds upon previous blocks in which proof-of-concept computer vision models were developed for plant phenotyping tasks such as object detection, segmentation, and measurement. The current focus is to productionise these models and deploy them to a cloud platform as part of an end-to-end MLOps pipeline. The context of the project relates to the HADES phenotyping system at the Netherlands Plant Eco-phenotyping Centre (NPEC), where large volumes of image data are used in combination with robotics to automate inoculation and analysis processes in plant breeding research.

The team is working in sprints using the Scrum methodology, supported by Azure DevOps. Each sprint is guided by defined deliverables and task estimations using the Fibonacci scale. This plan describes the project's approach sprint by sprint and outlines the key features under development.

Sprint 1 – Planning and Setup

The first sprint is dedicated to understanding the scope of the project, aligning on the MLOps concepts, and preparing the development environment. During this phase, we reviewed the deliverables, created a preliminary roadmap, and populated the backlog in Azure DevOps.

A GitHub repository was initialized with a branching strategy based on GitHub Flow. The team established rules for commit messages, pull requests, and branch naming conventions. Additionally, we set up virtual environments and package management tools to support isolated and reproducible development. The Python package structure was defined and aligned with documentation standards.

Sprint 2 – MVP Inference Application

The second sprint focuses on developing a minimal viable product (MVP) capable of inference. This includes a well-documented and installable Python package that supports logging, unit testing, and command-line interaction. The model is exposed via an API built with FastAPI and containerized using Docker. The application is tested locally and deployed to an on-premise server.

Documentation is generated using Sphinx, and test coverage is verified to ensure

maintainability. The MVP serves as the foundation for the deployment and integration work in future sprints.

Sprint 3 – Data Pipelines and Model Training

In the third sprint, the focus shifts to transitioning the data handling and training workflows into the Azure cloud environment. Data is uploaded and registered using the Azure ML SDK. Versioned datasets are created, and cloud-based environments are configured for reproducible training.

A model training pipeline is implemented, which includes data ingestion, preprocessing, training, evaluation, and conditional registration. A lightweight front-end using either Streamlit or Gradio is built to enable user interaction with the model. The front-end is containerized and prepared for deployment.

Sprint 4 – Deployment, Monitoring, and Retraining

Sprint four is centered on deploying the model to the cloud and implementing continuous integration and deployment (CI/CD) workflows. The team sets up deployment endpoints using Azure ML and enables logging and monitoring of prediction metrics. Performance indicators such as accuracy, latency, and confidence intervals are tracked.

An automated retraining pipeline is introduced, which can be triggered on a schedule or when new data is added. CI/CD scripts are created to manage the end-to-end flow from code changes to deployment. Monitoring dashboards are reviewed and validated for detecting model drift and degradation.

Sprint 5 – Testing, Evaluation, and Demo Preparation

The final sprint is focused on system testing, evaluation, and demonstration. The deployed application is tested with synthetic and real-world data to confirm performance and stability. User acceptance testing is conducted to ensure the front end and API behave as expected.

Final documentation is assembled, including user instructions, model information, and deployment steps. A project poster is created to support the demo day presentation. Peer reviews, sprint evidence, and final retrospectives are completed before project closure.

Review and Approval

This project plan and the corresponding roadmap were reviewed by the team during the first sprint. Adjustments were made based on team feedback, and deliverables were redistributed to balance the workload. The plan is now ready for final refinement and mentor approval.